

Real-Time Impact Point Prediction of Suborbital Rockets Using Kalman Filtering

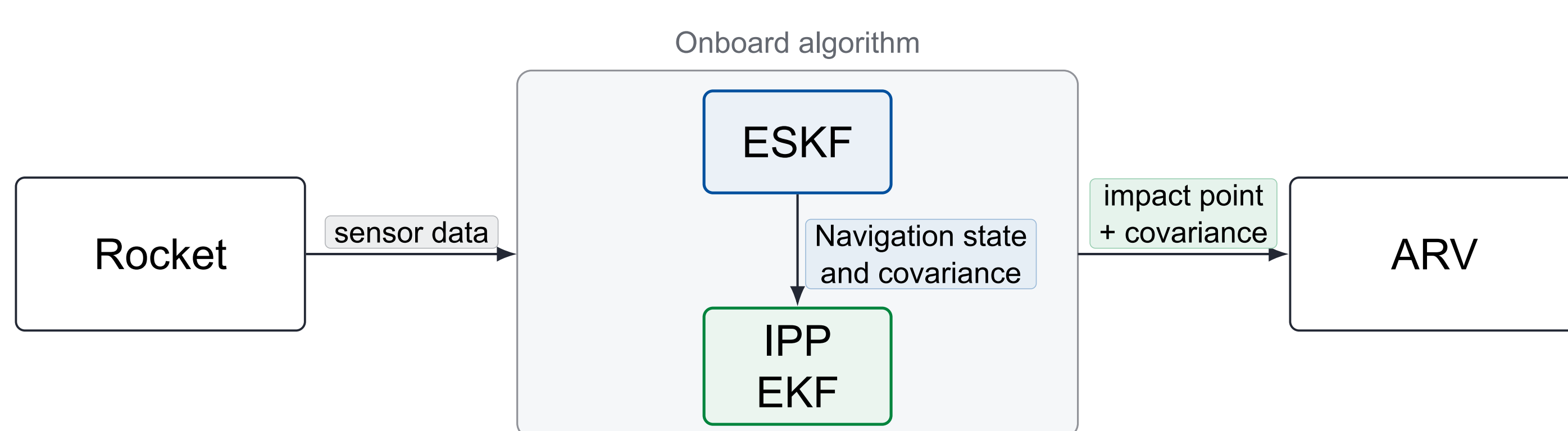
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Research Objective

Rocket launch rates are rising, making sustainable recovery of payloads and first stages increasingly important. Autonomous recovery vehicles (ARVs) address this by reducing cost, human exposure, and environmental footprint. This requires solving two coupled problems. The rocket's landing location must be predicted in real time so the ARV can be guided to that location.

Impact point prediction (IPP) is the real-time estimate of the rocket's landing location and uncertainty, updated continuously from onboard sensor data. An ARV steers to this estimate.



IPP Architecture

Navigation Filter

The ESKF maintains a continuous estimate of position, velocity, attitude, and sensor biases by combining inertial propagation with GNSS, barometer, and magnetometer aiding. The filter outputs a full state vector and covariance that are passed directly as the initial condition for the IPP filter.

Impact Point Prediction Filter

The IPP filter propagates a Cartesian state $\mathbf{x} = [\mathbf{p}^T, \mathbf{v}^T]^T$ in NED coordinates forward until altitude reaches sea level:

$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{g} - K_D \mathbf{v}_a + \mathbf{R}(\hat{\mathbf{q}}) \mathbf{T}_b, \quad K_D = \frac{\rho C_D A \|\mathbf{v}_a\|}{2m}$$

$\mathbf{v}_a$ airspeed vector, ρ air density, C_D drag coefficient, A reference area, m mass. $\mathbf{R}(\hat{\mathbf{q}})$ rotates the body-frame thrust $\mathbf{T}_b$ to NED using the ESKF attitude quaternion $\hat{\mathbf{q}}$.

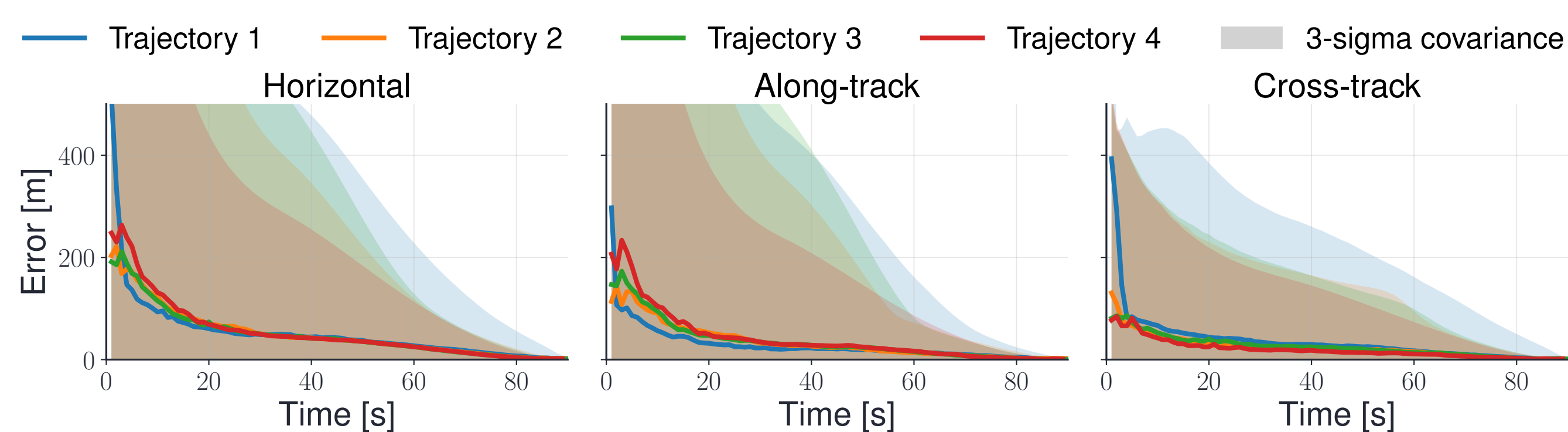
The equations are propagated forward until the predicted altitude reaches sea level, yielding a predicted impact point and propagated uncertainty ellipse updated in real time.

Component	Key design choice
ESKF	IMU propagation with GNSS, barometer, and magnetometer aiding
IPP	Forward propagation of position and velocity using EKF equations
Interface	IPP re-initialised from ESKF state and covariance at each GNSS update

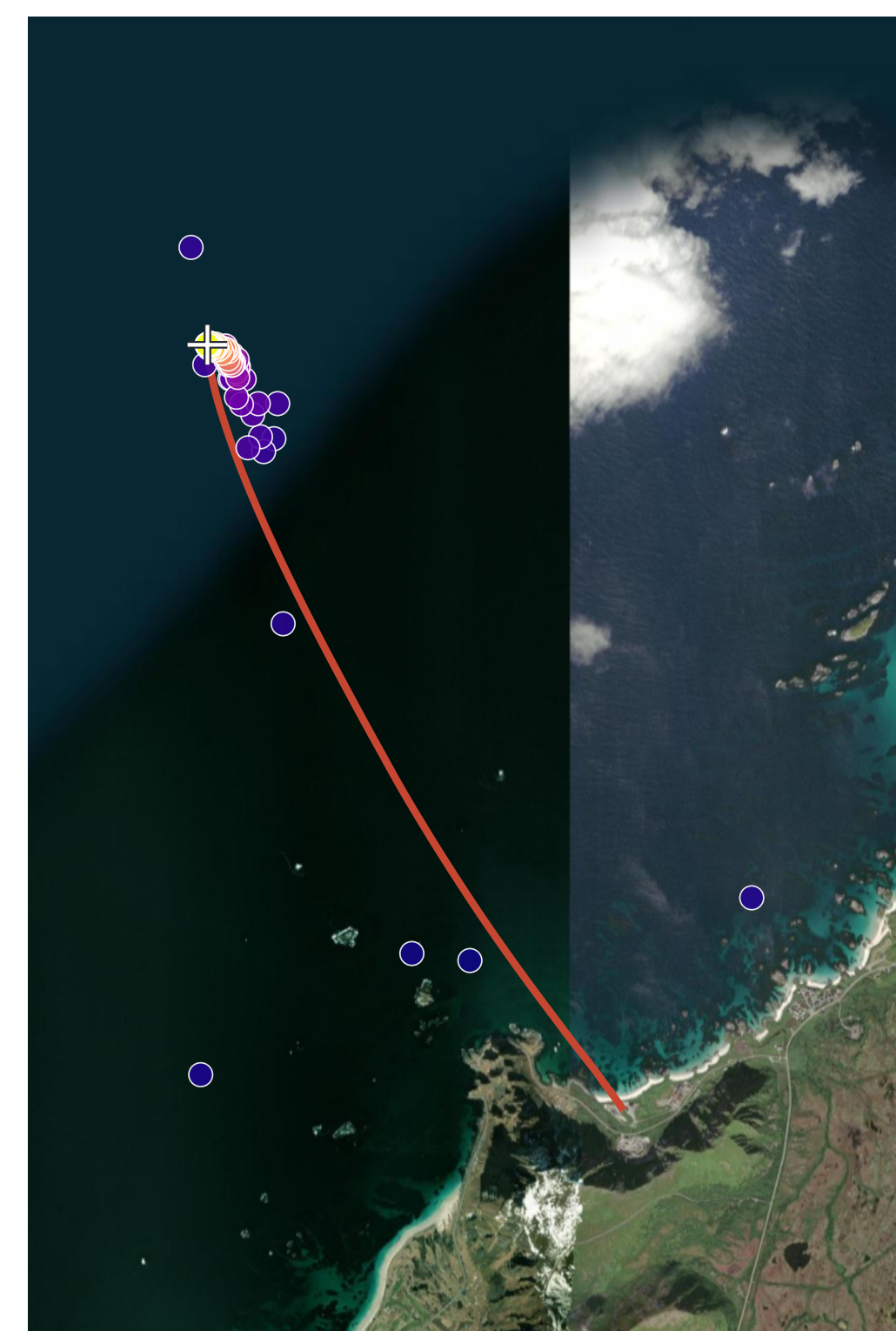
Simulation Performance

The Cartesian IPP architecture was evaluated on four simulated trajectories. For each trajectory, Monte Carlo runs were generated using perturbed wind estimates while keeping the vehicle model close to the simulated truth. These results provide a controlled reference without the hardware saturation observed in the flight data.

The Cartesian EKF model achieved a 45.3m mean RMSE across the four trajectories, indicating low prediction error in this controlled simulation setting. The plotted data represent the median error and 3-sigma covariance across 100 Monte Carlo wind realizations per trajectory.



Andøya Test Flight

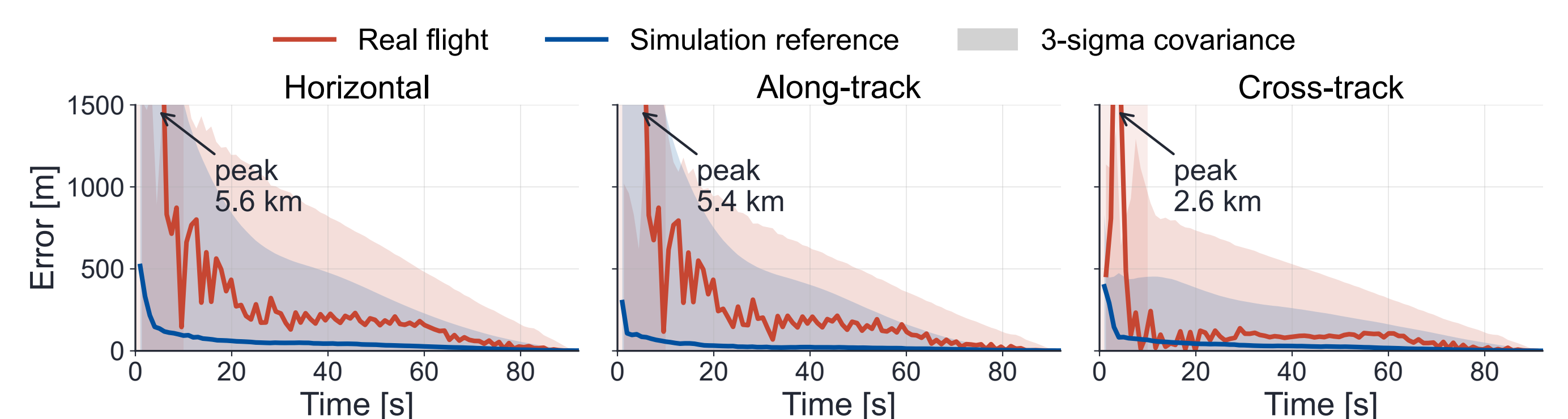


Andøya student rocket flight (Mongoose 98), 90 s flight to impact. The map shows the rocket ground track (red) and IPP predictions coloured by flight time. Late predictions converge toward impact.

The rocket carried an IMU, GNSS, and magnetometer. Barometer data was not available on this flight. Wind profiles came from a weather balloon launched 30–45 min before launch. The rocket impacted approximately 6.3 km from the launch pad.

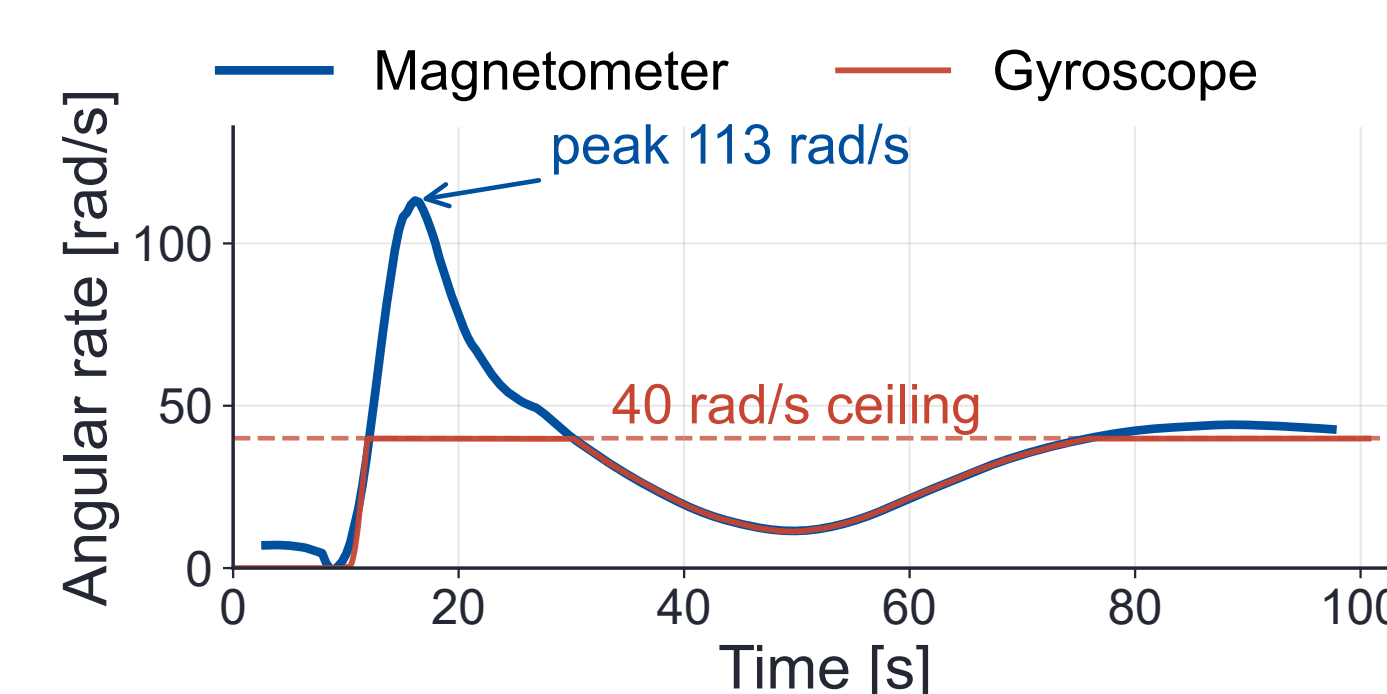
Burn phase outliers stem from initial yaw uncertainty and attitude instability caused by gyroscope saturation under high spin, possibly compounded by thrust model errors. Once ballistic, predictions converge toward the actual impact, marked by the crosshair.

Real Flight IPP Error



During the burn phase ($t \lesssim 10$ s) the prediction is corrupted by gyroscope and accelerometer saturation under high spin, vibration, GNSS error, and thrust/mass mismodelling. After burnout the error drops sharply and decays through to impact. The simulation reference is the Monte Carlo median for the corresponding simulated trajectory. Cross track error is of similar order to the reference, while larger along track error is consistent with unmodelled drag or mass effects. One flight is not sufficient for a general conclusion. Errors are reported against the GNSS reference.

Challenges & Outlook



Spin saturation. The gyroscope saturates at ± 40 rad/s through burn. The magnetometer recovers the true ≈ 113 rad/s peak. Mitigations include a gyroscope with larger angular rate range, IMU placement closer to the spin axis, and lower rocket spin.

- **Wind and drag uncertainty.** Wind estimation and drag identification remain open problems.
- **Known mass and thrust.** The IPP model treats vehicle mass and thrust as deterministic inputs. Propellant use and thrust variations introduce unmodelled uncertainty that limits accuracy before burnout. Estimating parameters in flight would address this.
- **Scaling to orbital recovery.** The same architecture could support first stage recovery, but would require propagation over a longer horizon, better atmosphere and aerodynamics models, and wider validation.

References

- [1] K. Gryte et al. Recovery of Rocket Payloads and First Stages Using Unmanned Vehicles. IEEE Aerospace Conference, 2025.
- [2] A. Meta. Impact Point Prediction of High-Power Rockets and Sounding Rockets. M.Sc. Thesis, NTNU, 2025.
- [3] E. K. Jensen. Rocket science: Catch space rockets using unmanned vehicles. M.Sc. Thesis, NTNU, 2026.